Model

Modeling is an important aspect of synthetic biology, where scientists and researchers test and evaluate the proposed aspect of a biological system prior to commencing laboratory work. Our modelling workflow primarily deals with the reengineering of the BtSULT enzyme for 5□ steroid specificity.

Enzyme Engineering

Wild type(WT) BtSULT is a sulfotransferase derived from Bacteroides thetaiotaomicron that solubilizes steroid molecules such as DHEA. BtSULT was selected after its biochemical characterization by Yao et. al (2022), demonstrating sulfotransferase activity on bile acids. However, the substrate selectivity profile of BtSULT reveals its inability to accept 5□ steroid molecules due to a kinked A/B fusion ring, such as 5□-LCA. This fact necessitates an engineering process to redesign the WT BtSULT active site to sulfonate 5□-LCA for PCS treatment

BtSULT was selected as the main engineering scaffold instead of its mammalian homolog, SULT2A1, considering BL21 expression bottlenecks. SULT2A1 crystal structures were utilized from UniProt as a structural template for active-site redesign.

The ideal reaction mechanism that indicates a working BtSULT mutant for wet-lab validation prioritizes sulfur-group transfer from PAPS to LCA 3-OH for production of sulfolithocholic acid, or sulfo-LCA.

According to Kakuta et al. (1998), their study of human estrogen sulfotransferase activity asserts sulfuryl transfer is only possible if the catalytic base contacts the substrate OH at hydrogen bond range, 2.7 to 3.1 Å. This is to be considered in our engineering process.

1. Structural Study of BtSULT

Although human SULT2A1 is well-studied and documented, BtSULT is not, requiring a study to identify candidates for a proposed catalytic base. Our early studies assumed BtSULT used histidine as a catalytic base, similar to His99 in human SULT2A1 (PDB: 1J99). Clustal Omega homology alignment tentatively suggested His215 as the closest histidine (Fig. 1). Extrapolating from the conserved PAPS binding motif, we also elect E136 as a catalytic base candidate. However, homology alignments were later considered unreliable and therefore inaccurate, partly due to the 15.2% sequence identity according to a Foldseek structural alignment (see below).

Foldseek Alignment Metrics

Metric	Value
Query	BtSULT WT (AlphaFold model)
Target	hSULT2A1 (PDB 1J99, DHEA-bound)
TM-score	0.4634
Sequence identity	15.2%
Alignment length	341 residues (BtSULT 58–365, SULT2A1 20–266)
lDDT	0.3757
E-value	8.33×10^{-4}
Bit score	78

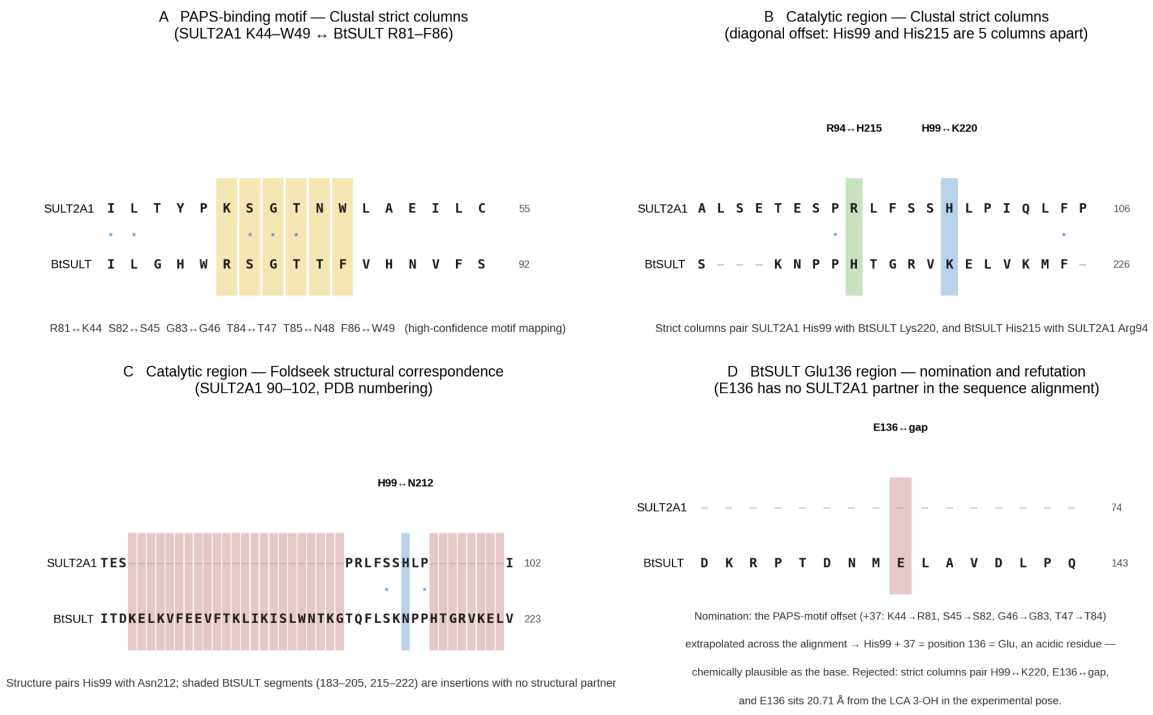


Figure 1. Clustal Omega Vs. Foldseek Alignment of BtSULT and SULT2A1

However, Boltz-2 co-folding poses, where BtSULT was complexed with PAPS and LCA, suggested E144 as the closest deprotonating residue to LCA 3-OH. Although E144 contacted the substrate OH outside of H-bond distance, E144 was still selected to account for co-folding overcompensating for the stability of the output pose, something that is not completely representative of an enzyme's tendency to change shape in its active site.

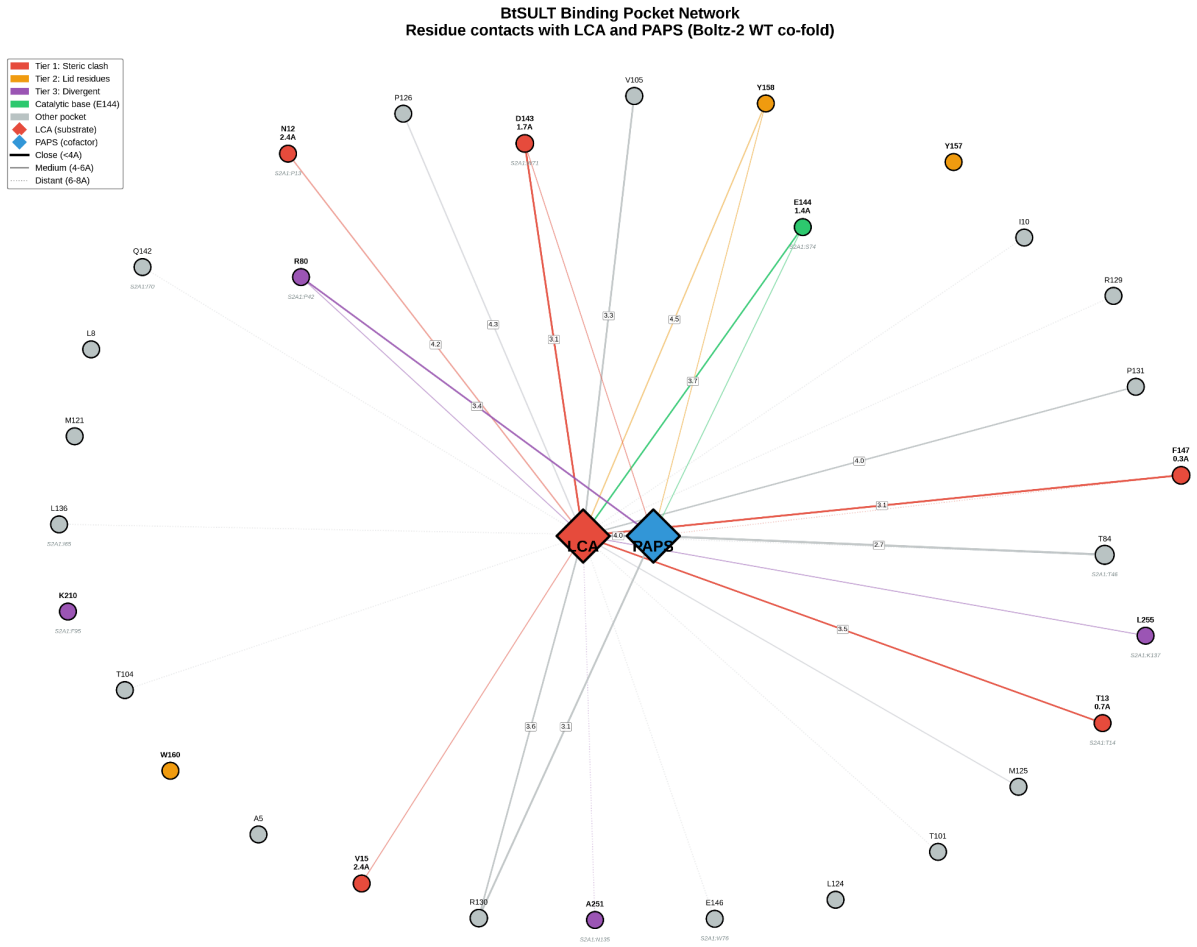


Figure 2. Boltz-2 LCA+PAPS Interaction Map

Additional candidates for the catalytic base were nominated from literature, on the condition that residues were within 5 Å of either LCA or PAPS. The candidates are summarized by Table 1.

Table 1: Catalytic base candidates

Candidate	**Route**	**Origin of Method**
His215	Homology + Sequence Alignment	SULT catalytic-His homology
Glu136	Sequence alignment	Clustal mapping of His99
Lys211	Literature	Tibbs catalytic-residue mapping (Tibbs et al.)

Candidate	Route	Origin of Method
Gln143	Steric Clash	Measured for completeness of study from Boltz-2 contact map
Tyr158	Literature	AKR1D1 Tyr58 (Chen et al.)
Glu144	Contact map	Boltz-2 contact analysis (closest protein contact to substrate OH)

2. Catalytic Distance Identification

In order to establish a baseline catalytic distance for which to discriminate between successfully and unsuccessfully engineered mutants, we are prompted to determine the optimal bond distance in SULT2A1-like enzymes.

The crystal structures of native SULT enzymes were analyzed for sulfuryl transfer and catalytic base distance, listed below. On average, sulfuryl transfer occurs at a distance of ~5 Å and catalytic bases are ~3 Å from the substrate OH (ref. Table 2).

Table 2: Catalytic Distance Inference from SULT Crystal Structures

Construct/PDB	Ligand(s) bound	Catalytic distance measured	Value
1J99	PAP + DHEA	His99 NE2 → substrate OH	3.15 Å
3F3Y	LCA + PAP	His99 → substrate OH	2.91 Å
3F3Y	LCA + PAP	substrate OH → PAP	5.44 Å

Additionally, in order to identify residues blocking 5β-LCA binding, LCA was posed onto wild-type(WT) BtSULT using SULT2A1-DHEA(1J99) as a template. After superimposing BtSULT onto 1J99, the DHEA pose was transferred to dock LCA with emphasis on the anchored 3-OH position. Residue-substrate distances were measured to the LCA pose and a selectivity filter for steric clashes was applied to residues using MolProbity, with an overlap of >0.4 Å classified as a steric clash. The main pocket residues are displayed in Fig. 3

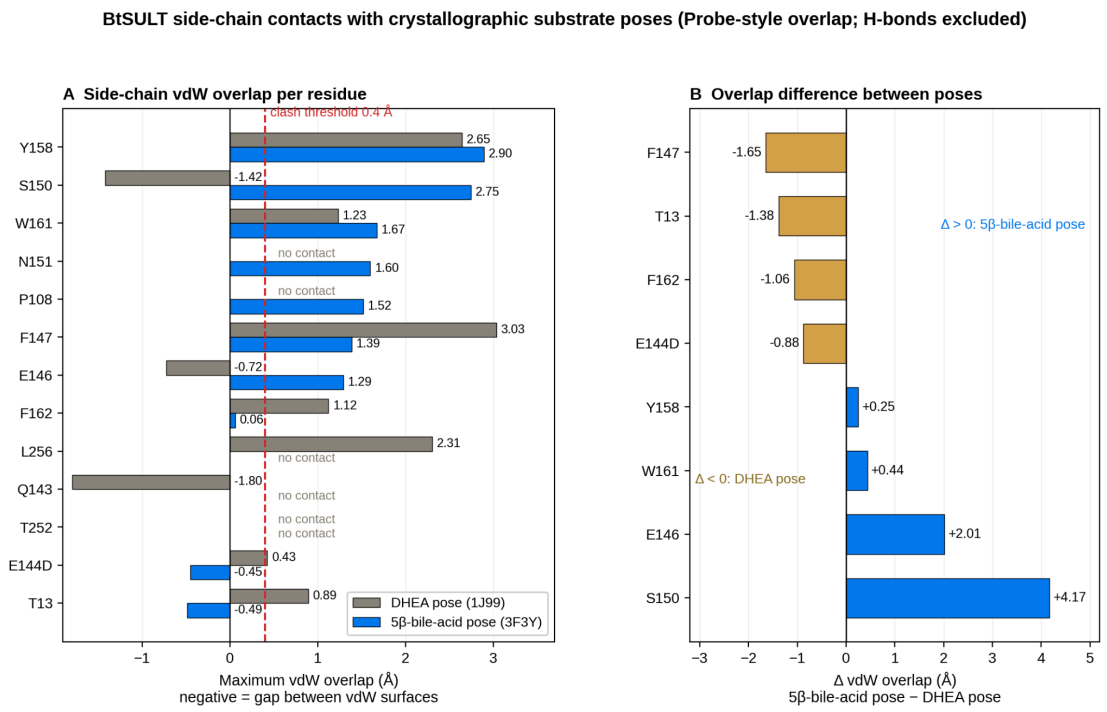


Figure 3. BtSULT MolProbity Absolute Clash Map

Although the greatest absolute clashes lie at Y158 and S150, studies of bile acid synthesis assert the importance of tyrosine in hydride transfer and brief analysis of the serine r-group dismissed it as an engineering focus due to its short, polar sidechain (Di Costanzo et al.). This shifted the focus to F147 as the principal steric clash, especially when considered as a selectivity marker for 5β steroids in human carbonic anhydrases (Brynda et al.).

3. Mutagenesis Screens

Round 0 of mutagenesis screening for the proposed enzyme prioritized a 6x-alanine scan, geared towards validating expression bottlenecks in-vitro prior to the final variant being reengineered. Steric clashes were changed to alanine as a proxy for catalytic activity in order to prioritize enzyme binding energy. This work was done prior to further structural studies, where Boltz-2 structural distances and review of literature confirmed the presence of mediating residues such as lysine, tyrosine, and glutamate as a divergent catalytic base to histidine in human AKR1D1. The mutations were as follows: E144A, W161A, K211A, F248A, F249A, T252A

3.1 Point Mutation Panel

Round 1 of enzyme engineering draws from the BtSULT structural study by introducing point mutations into the active site of BtSULT (Section 1). The proposed catalytic base of E144 was mutated to aspartate to relieve a steric clash due to its sidechain similarity to glutamate. Additionally, F147 was mutated to serine(F147S) to act as a smaller, stabilizing sidechain, testing if F147 was a selectivity marker for 5β steroids as it was in human carbonic anhydrases (Brynda et al.).

Serine was determined to be the best mutation at F147 after a saturation mutagenesis study was conducted, prioritizing sulfuryl transfer distance between the substrate and PAPS as well as OH-base distance as a proxy for bond formation. The results are displayed in Table 3.

Table 3: Saturation Mutagenesis of F147

Residue at 147	**Transfer distance (Å)**	**Base distance (Å)**
Ser	**4.26**	**2.46**
Ile	4.70	2.70
Thr	4.95	2.50
Ala	4.70	3.05
Phe (WT)	4.90	3.20

Residue at 147	Transfer distance (Å)	Base distance (Å)
Leu	5.50	3.40
Val (literature choice)	6.20	2.90
Gly	6.40	3.00

The serine mutation prioritized the smallest transfer and base distances, consistent with the formation of an H-bond, acting as an additional stabilizing residue to position the substrate more tightly into the catalytic pocket (Fig. 4) (Kakuta et al.).

Round 1 specifically searched for a switch between DHEA, a known BtSULT substrate, and LCA selectivity in WT and mutated BtSULT (Yao et al.). This was conducted by mutating two PAPS-motif residues into R236K and T252S, probing the PAPS side geometry while maintaining charge. His215 from then on was disproved as a catalytic base and mutations F147S seemed to decrease specificity for DHEA when the native catalytic base was above the measured 2.91 Å OH distance in PDB 3F3Y (See Section 1).

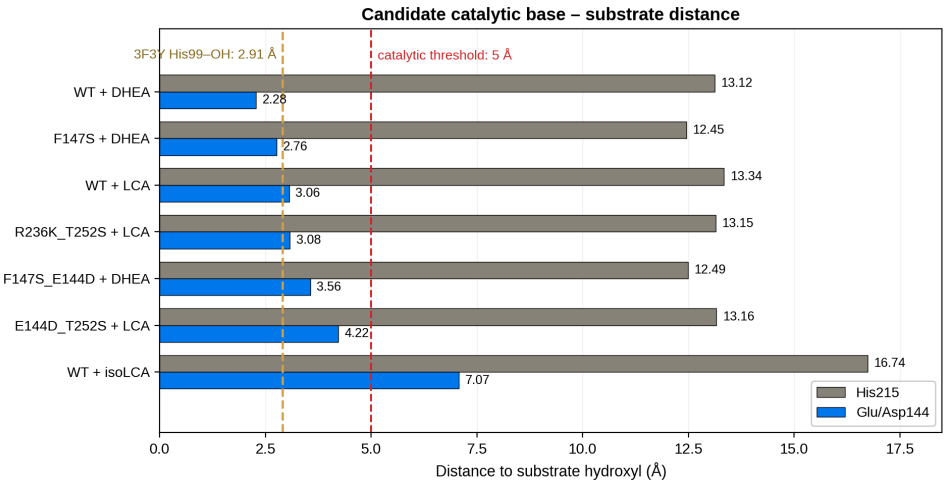


Figure 4. Round 1 point mutation distance to substrate hydroxyl

The testing strategy was strengthened, introducing new sets of mutations in Boltz-2 poses only against complexes of LCA and PAPS, considering OH distance, sulfuryl transfer, and Y158 mediation of sulfuryl transfer. Additional candidates mutated

E144 to histidine, utilizing SULT2A1's catalytic histidine as a replacement to the proposed E144 catalytic base. T13's earlier steric clash was relieved in further testing. These variants are shown in Table 4 and in Figure 4.

Table 4: Point Mutation Panel

Variant	**Mutations**	**Base → LCA 3-OH (Å)**	**LCA 3-OH → PAPS S (Å)**	**Y158-OH → LCA (Å)**	**ipTM**
6-Ala	6x Ala pocket scan	4.1	3.4	4.2	0.53
F1	E144H	3.5	4.4	7.0	0.55
F2	E144D	2.6	3.9	4.2	0.52
F3	F147A + E144H	3.7	7.9	3.9	0.48
F4	F147A + E144D + T13S	2.0	4.9	3.8	0.50

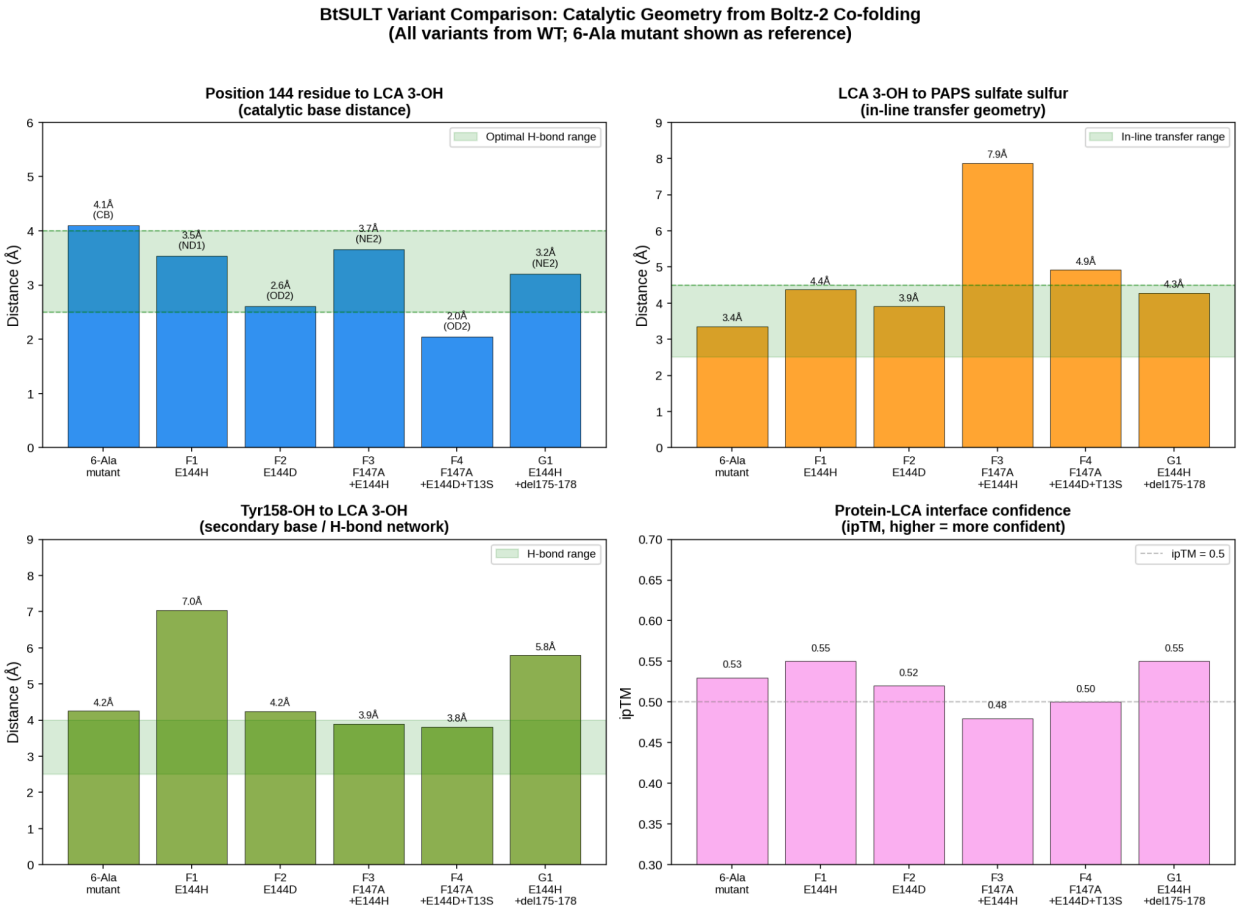


Figure 5. Boltz-2 substrate selectivity measurements

Fig. 5 confirms that F147S is a selectivity switch from DHEA to LCA for BtSULT in the 3/3 Boltz-2 ensemble testing, placing LCA within catalytic distance from 0/3 to 3/3 across each Boltz-2 ensemble.

3.2 Catalytic Loop Redesign + Grafting

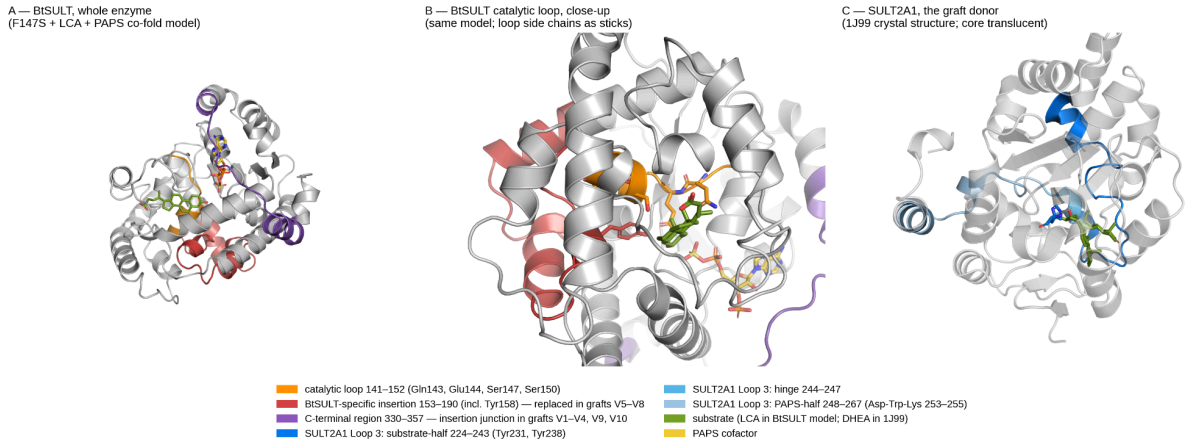


Figure 6. BtSULT Vs. SULT2A1 Catalytic Loop Analogies **A**: BtSULT + LCA + PAPS with specific insertion in red. **B**: Closeup of A. **C**: Donor chain on SULT2A1(blue) (Tibbs et al. 2015).

Round one of mutagenesis screening considered conserving catalytic geometry of BtSULT. However, round two considered redesigning the catalytic loop through a series of grafting experiments, utilizing a donor chain from SULT2A1, namely loop 3 (Fig. 6).

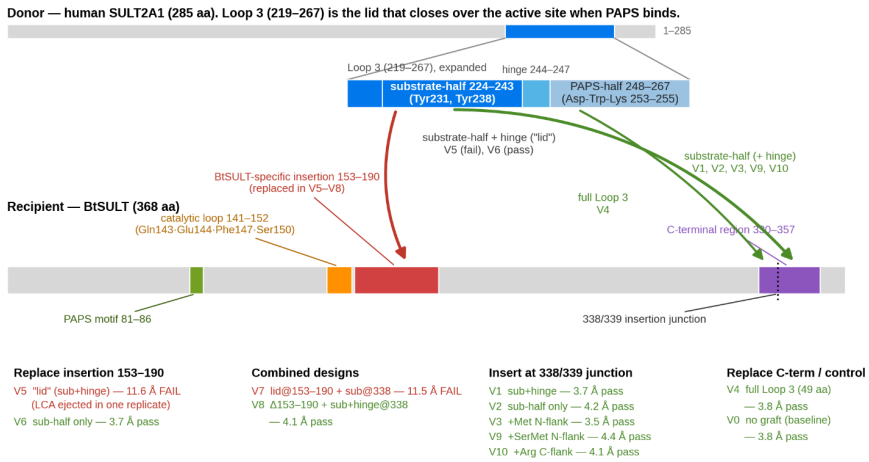


Figure 7. Graft design methodology, V0-V10. Top: SULT2A1 loop three divided into its three functional segments. Bottom: BtSULT regions as graft sites

SULT2A1 loop 3 was split into its three functional segments for candidate design: its substrate binding region, the loop's hinge region, and the PAPS binding motif.

These three segments were interchanged across 10 variants along two segments of BtSULT plus additional flanking residues. Residues 153-190 hosted a BtSULT specific insertion, revealed by the initial ClustalOmega alignment and its proximity to the proposed E144 catalytic region. The substrate-binding region of SULT2A1 was grafted to this region. The C-terminus of BtSULT, aligning with SULT2A1's catalytic loop, hosted either a grafted substrate-binding motif, a PAPS-binding motif, or both (Fig. 7).

Additionally, a 3X alanine background was added to each variant, following clash analysis of the 6X alanine background from Round 0 and Round 1. Glu144 and Lys211 had been assigned as key pocket residues and were therefore not mutated. Similarly, Trp161 was left unchanged due to its location in the 153-190 BtSULT insertion. This controlled our analysis of variants with grafts in the 153-190 region instead of adding a confounding variable (Fig. 8).

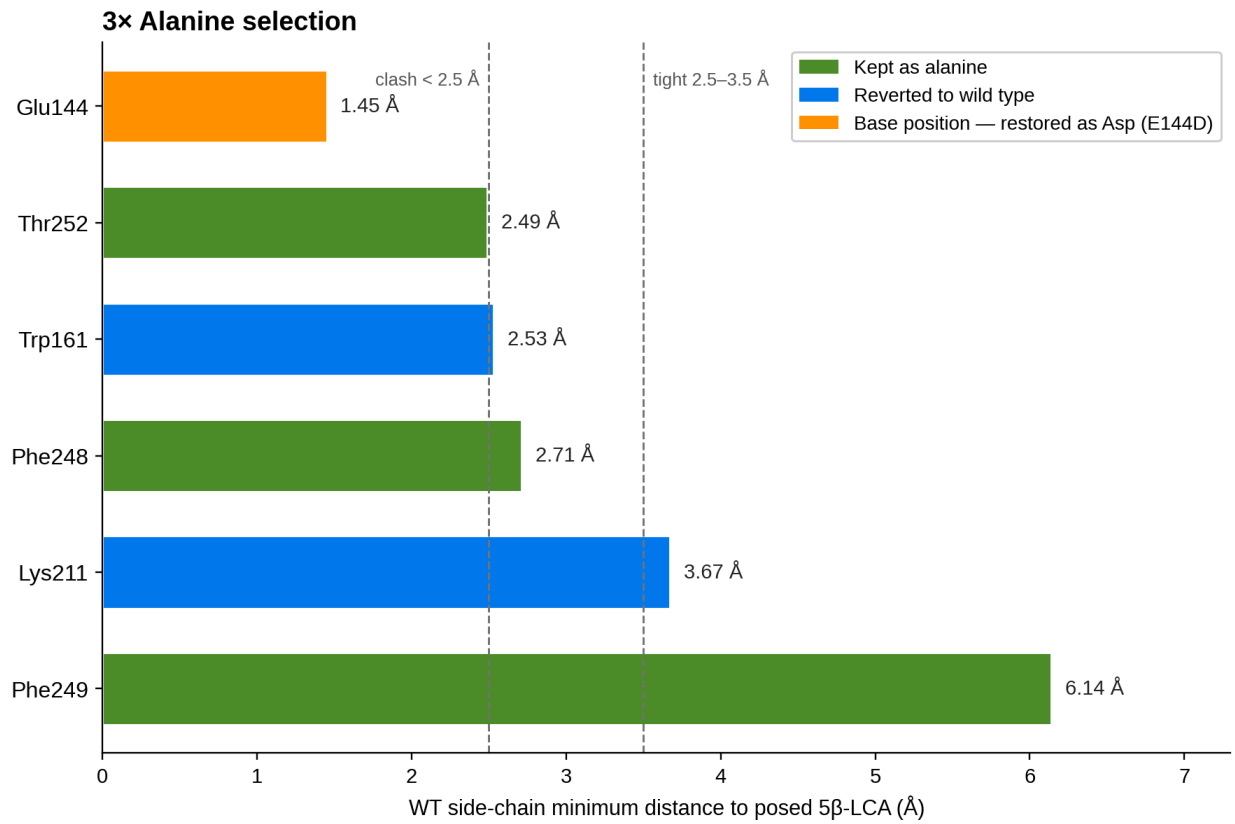


Figure 8. 3X Alanine Selection

The variants are summarized and analyzed in Table 5.

Table 5: Loop Graft Summary

Variant	**Design**	**Hypothesis**	**Catbase (Å)**	**Transfer (Å)**	**Comments**
V0	No graft (E144D + 3×Ala background)	Baseline control	3.70	3.81	–
V1	Substrate-half + hinge → 338/339	The substrate-contacting piece is enough	4.25	3.66	Junction accepts inserts
V2	Substrate-half only → 338/339	Is the hinge needed?	2.92	4.19	Hinge assists but isn't significant.

Variant	Design	Hypothesis	Catbase (Å)	Transfer (Å)	Comments
					Catbase improves but transfer alignment doesn't worsens
V3	+ Met N-flank	A flanking residue eases the junction scar	3.92	3.54	Best junction insert
V4	Full Loop 3 (49 aa) replaces 330–357	Utilize the whole lid	3.32	3.83	Full loop works at the C-terminal site
V5	Substrate-half + hinge replaces 153–190	Place the lid in the insertion region	9.78	11.61	Failed Design
V6	Substrate-half only replaces 153–190	Same site, shorter lid length	2.71	3.67	Lid-length may be the issue, learned from V5
V7	Lid at 153–190 + substrate-half at 338	Utilize both sites	9.22	11.53	Failed Design
V8	Delete 153–190 + substrate-half+ hinge at 338	Increase SULT2A1 homology, insert SULT2A1 components	2.21	4.08	Structure is deletion tolerant
V9	+ Ser-Met N-flank	Is more flanking better?	3.48	4.44	More flanking is not necessary

Variant	Design	Hypothesis	Catbase (Å)	Transfer (Å)	Comments
V10	+ Arg C-flank	Opposite flank	4.01	4.05	Junction tolerates either side
graft_Asp144	V5 without alanines. Base is replaced with Asp144	Re-test the lid without alanine background	2.46	4.54	Alanine background failed V5.
graft_Glu144	V4 without alanine background. Same catalytic base	Re-test the lid without alanine background.	2.87	5.74	The original catbase doesn't allow transfer in graft. Is grafting still viable?

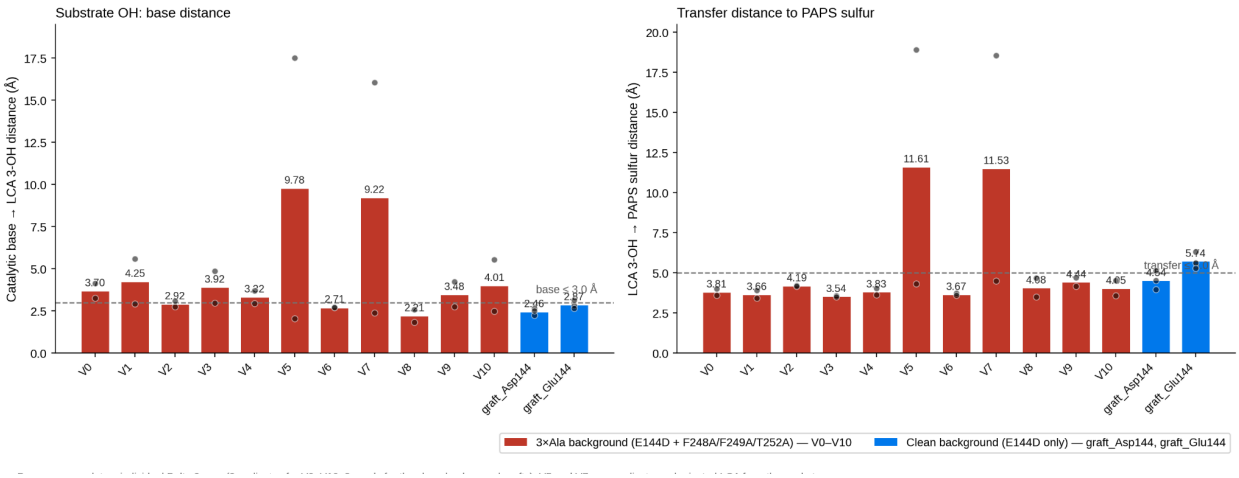


Figure 9. Full Graft Screen Summary

With exception of V5 and V7, which were retried without the 3X-Ala background, all variants maintained transfer distance to PAPS sulfur. Standout variants included V2, V6, V8, and graft_Asp144. However, considering the diversity of the employed grafting techniques and a lack of variability between variants may render the grafting method ineffective at producing unique results (Fig. 9). This is in light of sulfotransferase reaction kinetics, undergoing a conformational shape change

once PAPS initially binds, allowing the substrate to enter the active site (Tibbs et al.).

Furthermore, V2 improves catalytic base distance when the native hinge region is maintained at the cost of transfer distance, and was therefore rejected due to inconclusive results. Analysis of V6 supported a grafted substrate lid that was functional, leading to the V5 redesign and discovery of graft_Asp144 without alanine background. V8 informed us that the BtSULT insertion was deletion-tolerant, although highly experimental, causing V8 to be rejected for the wet lab pipeline.

3.3 ProteinMPNN Redesign + Chimeras

Round 3 of mutagenesis introduced the use of ProteinMPNN as a bulk method of introducing mutations intended to improve the viability of the enzyme in-vitro and catalytic activity with LCA. 10 trials were run, applying 20-27 mutations based on their ProteinMPNN score. Table 6 summarizes the ProteinMPNN runs with added baselines from Rounds 1 and 2 for reference.

Table 6: ProteinMPNN Summary Table

Design	**Background**	**Mutations**	**Base → LCA (Å)**	**LCA → PAPS (Å)**	**ipTM**	**P/F**
F2_E144D baseline	—	1	2.61	3.91	0.962	Reference
graft_Asp144	F2_E144D	lid graft + E144D	2.65	5.14	0.961	Reference
graft_Glu144	WT	lid graft	2.83	5.61	0.960	Reference
MPNN_F2_2	F2_E144D	26	2.04	5.85	0.944	Pass?
MPNN_F2_4	F2_E144D	26	5.82	11.96	0.926	Fail

Design	Background	Mutations	Base → LCA (Å)	LCA → PAPS (Å)	ipTM	P/F
MPNN_F2_1	F2_E144D	27	12.59	14.16	0.937	Fail (substrate detached)
MPNN_F2_7	F2_E144D	25	12.09	18.24	0.911	Fail
MPNN_F2_8	F2_E144D	27	12.34	16.78	0.908	Fail
MPNN_F2_3	F2_E144D	26	13.82	17.07	0.931	Fail
MPNN_F2_6	F2_E144D	25	14.03	18.37	0.920	Fail
MPNN_WT_3	WT	22	13.20	17.13	0.947	Fail
MPNN_WT_2	WT	23	13.87	18.00	0.959	Fail
MPNN_WT_4	WT	21	14.23	17.48	0.955	Fail
MPNN_WT_1	WT	21	15.37	19.39	0.952	Fail

Only MPNN_F2_2 showed an initial pass for catalytic geometry. However, a Boltz-2 ensemble was run to confirm its viability as a candidate (Table 7).

Table 7: 3-seed Boltz-2 Run of MPNN_F2_2

Seed	Base → LCA (Å)	LCA → PAPS (Å)
1	2.04	5.85
2	11.74	13.78
3	13.26	16.65

Only 1 of the three seeds displayed catalytic geometry. Therefore, the ProteinMPNN design method can be ruled out as viable.

An alternate path of AI-guided design was utilized through generation of a chimera between BtSULT and SULT2A1 using RFdiffusion. 50 backbones were generated from 0-49. In order to sample diversity of the backbone design, chimeras 0, 25, and 49 were selected to evaluate the robustness of the design method (Table 8).

Table 8: Chimera Summary Table

Construct	**Base → LCA (Å)**	**LCA → PAPS (Å)**
chimera_25 (best chimera)	3.4	9.33
chimera_0	13.7	17.35
chimera_49	15.9	18.38

Review of the generated chimeras clearly indicates that chimera generation is not a competitive method for mutagenesis. Although OpenMM MD was considered, a conversation with a mentor pushed us to conclude our enzyme engineering process in order to finalize candidates for experimental validation.

4. Summary and Wet-Lab Candidates

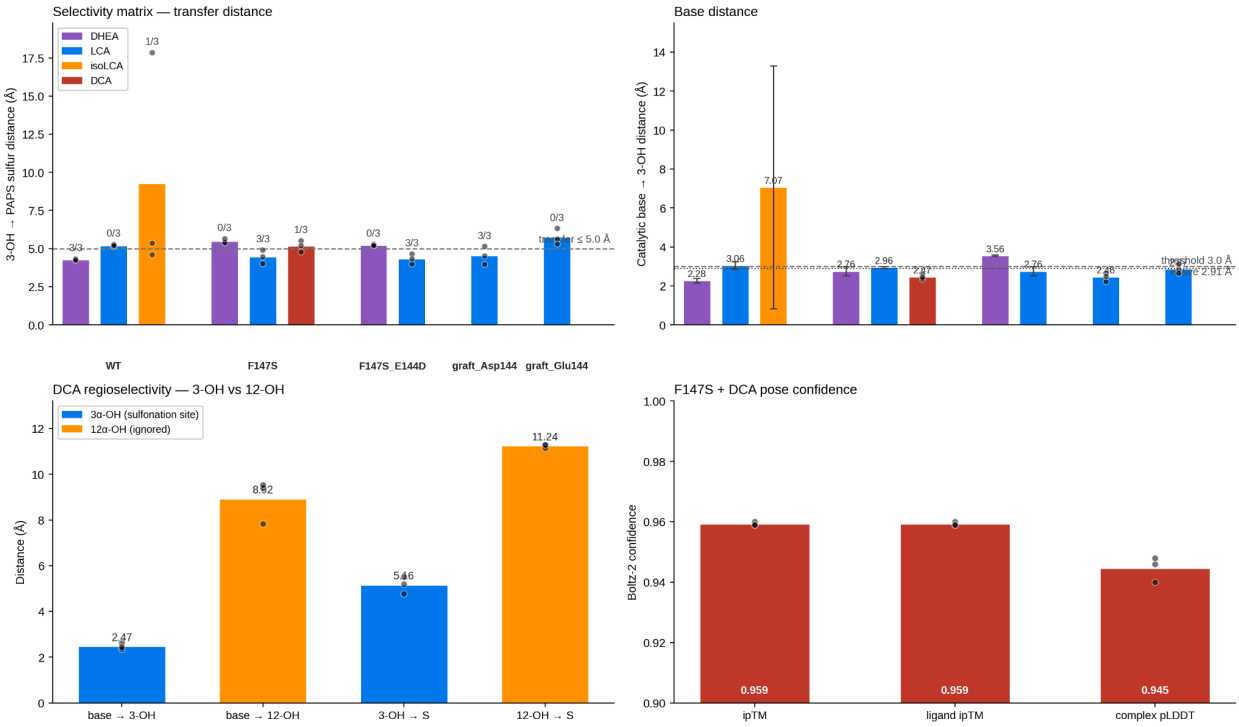


Figure 10. Best Candidates Full Substrate Ensemble

The best candidates were elected from each round: F147S, F147S_E144D, and graft_Asp144 and graft_Glu144. Structures were run in a 3-seed Boltz-2 ensemble against DHEA, the native WT substrate, LCA, DCA, and isoLCA (Yao et al. 2022).

Although graft_Asp144 demonstrated the highest specificity for LCA in terms of maintaining catalytic distances for sulfuryl transfer, the substitution of the proposed catalytic base, Glu144, deemed this technique too experimental for a limited wet lab pipeline. Furthermore, F147S demonstrated the strongest specificity towards LCA. Although above the threshold for DCA catalysis, the F147S variant preserves Glu144. In addition to F147S, F147S_E144D was also chosen for in-vitro testing in order to validate the Glu→Asp hypothesis as an alternative catalytic base (Fig. 10).

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